

Tutorial Sheet - IV

Q.1. Let X be a finite dimensional vector space.
A linear map $M \in \mathcal{K}(X, X)$ is similar to $N \in \mathcal{K}(X, X)$ if $\exists$ an invertible element $S \in \mathcal{K}(X, X)$ such that
$$M = SNS^{-1}.$$

- (i) Show that similarity is an equivalence relation.
- (ii) If either M or N in $\mathcal{K}(X, X)$ is invertible, then MN and NM are similar.
- (iii) If A and B are similar show that $\dim(R_A) = \dim(R_B)$ and $\dim(N_A) = \dim(N_B)$.

Q2. Let X be a vector space over F with basis $\alpha = \{x_1, \dots, x_n\}$. For each $1 \leq i \leq n$, define a linear functional $l_i: X \rightarrow F$ by taking linear extensions of the following: $l_i(x_j) = \delta_{ij}$, $\forall j = 1, \dots, n$
where $\delta_{ij} = \begin{cases} 1_F, & \text{if } i=j \\ 0, & \text{if } i \neq j \end{cases}$.

Show that $\ell = \{l_1, \dots, l_n\}$ is a basis for X^* .
Such a basis ℓ is called the dual basis of α .

Q3: Let X be an n -dimensional vector space over F , and let ℓ be a non-zero linear functional on X .
(i) What is the dimension of N_ℓ ?
(ii) If f is a linear functional on X ,
show that $f = \alpha\ell$ for some $\alpha \in F$ if and only if

$$N_g \subseteq N_f.$$

(iii) Let V be a subspace of X with $\dim(V) = n-1$. (Such a subspace is called a hyperspace). Find a linear functional $\gamma: X \rightarrow F$ such that

$$N_\gamma = V.$$

(iv) Find a linear functional γ on $\mathbb{R}^2$ such that N_γ is the line $ax + by = 0$, $a, b \in \mathbb{R}$.

Q4. Let T be a linear map on $\mathbb{R}^2$ defined by

$$T(x_1, x_2) = (-x_2, x_1).$$

(i) Find the matrix representation of T in the standard ordered basis of $\mathbb{R}^2$.

(ii) Find the matrix representation of T in the ordered basis $\{(1, 2), (1, -1)\}$.